

Q1. Debug the Broken Dockerfile

Answer

Bug 1: Incorrect Sequence of Execution

Incorrect Code:

None

```
USER appuser
```

```
RUN useradd -m appuser
```

Explanation:

Found the bug on line 3 and 4. Here the user is not created first its used first. The user needs to be created first then used. We are using the “appuser” but its doesn’t exist.

Correct Code:

None

```
RUN useradd -m appuser
```

```
USER appuser
```

Complete Correct Code - Dockerfile

Q2. Debug the Broken Compose File

```
GNU nano 7.2                                compose.yaml *
```

```
services:
  app:
    image: myapp:1.0
    ports:
      -8080:3000
    environment:
      -APP_ENV=production
    depends_on:
      - database

  db:
    image: postgres
    environment:
      - POSTGRES_PASSWORD=secret
    volumes:
      - dbdata:/var/lib/postgresql/data
```

Bugs Identified

Bug 1: Wrong Syntax (Missing Space)

The problem is that there should be a “space” after the dash.

Wrong syntax:

None

-8080

-APP_ENV=production

Correct syntax:

None

- 8080
- APP_ENV=production

Bug 2: Mismatch Service Name

```
depends_on:
  - database

db:
  image: postgres
  environment:
    - POSTGRES_PASSWORD=secret
```

`depends_on` says `database`, but the actual service is named `db`.

Correct Configuration:

```
depends_on:
  - db

db:
  image: postgres
  environment:
    - POSTGRES_PASSWORD=secret
```

Bug 3: Volume Not Declared

The file uses a volume called `dbdata`, but `dbdata` is not declared in the Compose file.

Before:

```
- POSTGRES_PASSWORD=secret
volumes:
  - dbdata:/var/lib/postgresql/data
```

Correct Declaration:

```
- POSTGRES_PASSWORD=secret
volumes:
  - dbdata:/var/lib/postgresql/data

volumes:
  dbdata:
```

Testing

Complete Correct Compose File

```
GNU nano 7.2                                compose.yaml *
```

```
services:
  app:
    image: myapp:1.0
    ports:
      - 8080:3000
    environment:
      - APP_ENV=production
    depends_on:
      - db

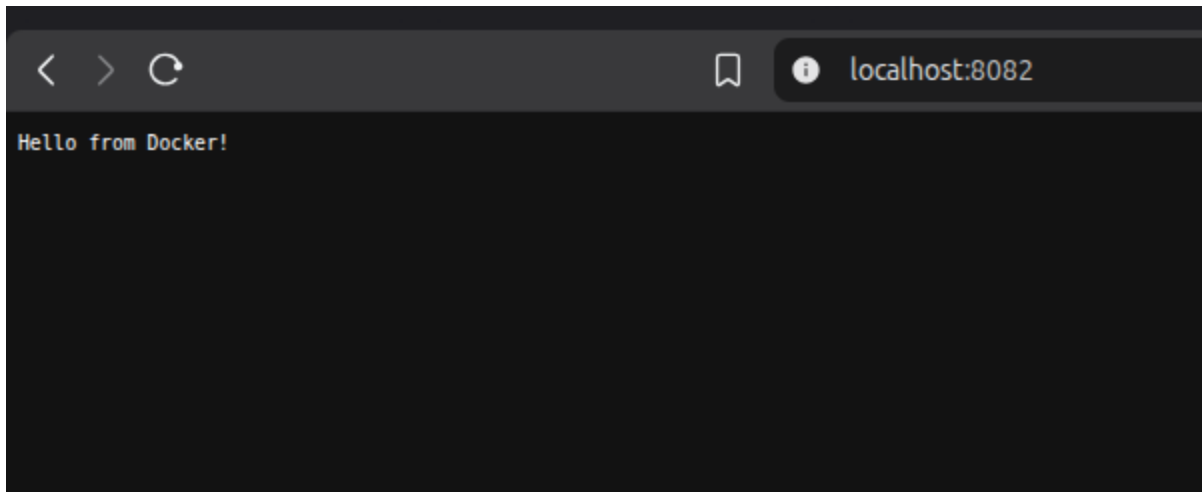
  db:
    image: postgres
    environment:
      - POSTGRES_PASSWORD=secret
    volumes:
      - dbdata:/var/lib/postgresql/data

volumes:
  dbdata:
```

```
rushi@rushi:~/Docker_assign$ docker compose up -d
[+] up 19/19
✓ Image postgres:17 Pulled 39.4s
✓ Network docker_assign_default Created 0.1s
✓ Volume docker_assign_dbdata Created 0.0s
✓ Container docker_assign-db-1 Started 0.6s
✓ Container docker_assign-app-1 Started 0.7s
```

```
rushi@rushi:~/Docker_assign$ docker ps
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAMES
0adc35232866	myapp:1.0	"docker-entrypoint.s..."	24 seconds ago	Up 23 seconds	0.0.0.0:8082->3000/tcp, :::8082->3000/tcp	docker_assign-app-1
88741cb324a1	postgres:17	"docker-entrypoint.s..."	24 seconds ago	Up 23 seconds	5432/tcp	docker_assign-db-1



Q3. Build a Production-Ready Stack

File Structure

```
rushi@rushi:~/Docker_assign$ tree -a
.
├── compose.yaml
├── Dockerfile
├── .env
├── package.json
└── server.js

1 directory, 5 files
```

Package Configuration (package.json)

```
GNU nano 7.2 package.json
{
  "name": "docker_lec",
  "version": "1.0.0",
  "main": "index.js",
  "scripts": {
    "test": "echo \\\"Error: no test specified\\\" && exit 1"
  },
  "keywords": [],
  "author": "",
  "license": "ISC",
  "description": ""
}
```

Server Code (server.js)

```
GNU nano 7.2 server.js
const http = require("http");

const server = http.createServer((req, res) => {
  res.writeHead(200, {
    "Content-Type": "text/plain"
  });

  res.end("Hello from Docker!\n");
});

server.listen(3000, "0.0.0.0", () => {
  console.log("Server running on port 3000");
});
```

Dockerfile

```
GNU nano 7.2 Dockerfile *
FROM node:20-alpine

RUN apk add --no-cache curl

WORKDIR /app

COPY package.json .

RUN npm install

COPY . .

USER node

EXPOSE 3000

HEALTHCHECK --interval=30s --timeout=3s \
  CMD curl -f http://localhost:3000/ || exit 1

CMD ["node", "server.js"]
```

Docker Compose Configuration (compose.yaml)

```
GNU nano 7.2                                     compose.yaml
services:
  app:
    build: .
    ports:
      - "8082:3000"

    restart: unless-stopped

    environment:
      - APP_ENV=production
      - DB_HOST=db
      - DB_PORT=5432

    depends_on:
      - db

    mem_limit: 256m

  db:
    image: postgres
    environment:
      - POSTGRES_PASSWORD=${POSTGRES_PASSWORD}

    volumes:
      - dbdata:/var/lib/postgresql/data

volumes:
  dbdata:
```

Build and Start Stack

1. Build and Run Containers

Command:

```
Shell
docker compose up --build -d
```

```

rushi@rushi:~/Docker_assign$ docker compose up --build -d
[+] Building 13.8s (14/14) FINISHED
=> [internal] load local bake definitions                                0.0s
=> => reading from stdin 5028                                           0.0s
=> [internal] load build definition from Dockerfile                     0.0s
=> => transferring dockerfile: 293B                                     0.0s
=> [internal] load metadata for docker.io/library/node:20-alpine       2.5s
=> [auth] library/node:pull token for registry-1.docker.io             0.0s
=> [internal] load .dockerignore                                         0.0s
=> => transferring context: 2B                                           0.0s
=> [1/6] FROM docker.io/library/node:20-alpine@sha256:fb4cd12c85ee03686f6af5362a0b0d56d50c58a04632e6c0fb8363f609372293 5.7s
=> => resolve docker.io/library/node:20-alpine@sha256:fb4cd12c85ee03686f6af5362a0b0d56d50c58a04632e6c0fb8363f609372293 0.0s
=> => sha256:fb4cd12c85ee03686f6af5362a0b0d56d50c58a04632e6c0fb8363f609372293 7.67kB / 7.67kB 0.0s
=> => sha256:afdf98210b07b586eb71fa22ba2e432e058e4cd1304d31ed60880755b8c865fb 1.72kB / 1.72kB 0.0s
=> => sha256:11cedc39e663e7c5d5cb9cc77a461a0d2adc25537b94e6831a6108f09cb2001b 6.52kB / 6.52kB 0.0s
=> => sha256:6a0ac1617861a677b045b7ff88545213ec31c0ff08763195a70a4a5adda577bb 3.86MB / 3.86MB 0.6s
=> => sha256:4feea04c154301db6f4a496efa397b3db96603b1c009c797cfdde77bea8b3287 43.23MB / 43.23MB 3.9s
=> => sha256:b2cbbfe903b0821005780971ddc5892edcc4ce74c5a48d82e1d2b382edac3122 1.26MB / 1.26MB 1.6s
=> => extracting sha256:6a0ac1617861a677b045b7ff88545213ec31c0ff08763195a70a4a5adda577bb 0.1s
=> => sha256:fff4e2c1b189bf87d63ad8bd07f7f4eb288d6f2b6a07a8bb44c60e8c075d2096 445B / 445B 0.9s
=> => extracting sha256:4feea04c154301db6f4a496efa397b3db96603b1c009c797cfdde77bea8b3287 1.4s
=> => extracting sha256:b2cbbfe903b0821005780971ddc5892edcc4ce74c5a48d82e1d2b382edac3122 0.0s
=> => extracting sha256:fff4e2c1b189bf87d63ad8bd07f7f4eb288d6f2b6a07a8bb44c60e8c075d2096 0.0s
=> [internal] load build context                                         0.0s
=> => transferring context: 1.37kB                                       0.0s
=> [2/6] RUN apk add --no-cache curl                                   1.4s
=> [3/6] WORKDIR /app                                                  0.1s
=> [4/6] COPY package.json .                                           0.0s
=> [5/6] RUN npm install                                               3.4s
=> [6/6] COPY . .                                                       0.1s
=> exporting to image                                                  0.1s
=> => exporting layers                                                  0.1s
=> => writing image sha256:d51c1e9c94f5eb3faf092601dda4cf63aeec3f49fa768b93e1c29c8dc70d5336 0.0s
=> => naming to docker.io/library/docker_assign-app                   0.0s
=> => resolving provenance for metadata file                             0.0s
[+] up 3/3
✓ Image docker_assign-app Built 13.9s
✓ Container docker_assign-db-1 Started 11.2s
✓ Container docker_assign-app-1 Started 11.3s
rushi@rushi:~/Docker_assign$

```

2. Check Container Status and Health

Command:

Shell

docker compose ps

```

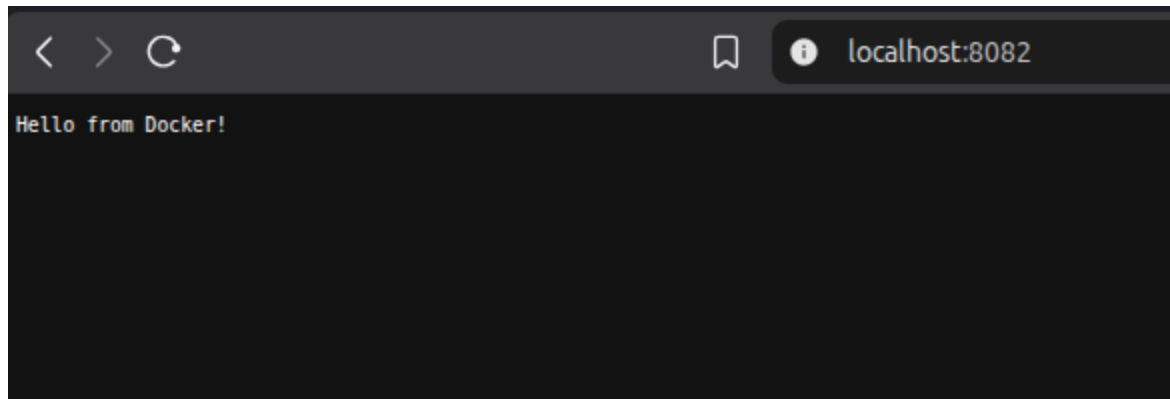
rushi@rushi:~/Docker_assign$ docker compose ps -a
NAME                IMAGE                COMMAND                SERVICE    CREATED          STATUS              PORTS
docker_assign-app-1  docker_assign-app    "docker-entrypoint.s..." app         About a minute ago Up About a minute (healthy) 0.0.0.0:8082->3000/tcp,
000/tcp
docker_assign-db-1   postgres            "docker-entrypoint.s..." db          About a minute ago Exited (1) About a minute ago
rushi@rushi:~/Docker_assign$

```

✓ **Passed the Health Check**

Testing Application Endpoint

Status: Success!



Verification of Non-Root User

Configured non-root user in Dockerfile using: **USER node**

```
rushi@rushi:~/Docker_assign$ docker compose exec app whoami  
node  
rushi@rushi:~/Docker_assign$
```